Fiscal Year:	FY 2024	**Task Last Updated:**	FY 10/29/2024
PI Name:	Stankovic, Aleksandra Ph.D.		
Project Title:	Quantification of Response to Virtual Reality-based Sensory Stimulation for Relaxation and Therapeutic Release in ICE		
Division Name:	Human Research		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:			
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	(1) **HFBP**:Human Factors & Behavioral Performance (IRP Rev H)		
Human Research Program Risks:	(1) **BMed**:Risk of Adverse Cognitive or Behavioral Conditions and Psychiatric Disorders (2) **HSIA**:Risk of Adverse Outcomes Due to Inadequate Human Systems Integration Architecture		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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City:	Charlestown	**State:**	MA
Zip Code:	02129-2020	**Congressional District:**	7
Comments:			
Project Type:	Ground	**Solicitation / Funding Source:**	2017-2018 HERO 80JSC017N0001-HHCHFBP: Human Health Countermeasures, Human Factors, Behavioral Performance. Appendix D
Start Date:	09/09/2020	**End Date:**	09/08/2025
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA JSC
Contact Monitor:	Whitmire, Alexandra	**Contact Phone:**	
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Flight Program:			
Flight Assignment:	NOTE: End date changed to 09/08/2025 per L. Juliette/HFBP (Ed., 6/4/24). NOTE: End date changed to 09/08/2024 per L. Juliette/HFBP (Ed., 8/15/23).		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Buckey, Jay M.D. (Dartmouth College) Bovard, Pooja Ph.D. (Charles Stark Draper Laboratory Inc) Strangman, Gary (Massachusetts General Hospital/Harvard Medical School)		
Grant/Contract No.:	80NSSC20K1852		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	The environmental conditions of prolonged spaceflight missions pose medical and psychological risks for astronauts. As identified by NASA Human Research Program (HRP), long duration exposure to an isolated, confined, and extreme (ICE) environment contributes to the risk of adverse cognitive or behavioral events which may compromise mission safety and success. Previous work has suggested a link between the reduced sensory stimulation associated with such environments and a loss of pleasure, satisfaction, and engagement ([1], [2], [3]). Effective countermeasures are necessary to promote individual behavioral health and performance by providing increased sensory stimulation, offering novelty, preventing boredom, reducing stress, and increasing attention. This study investigates the application of Virtual Reality (VR) stimulation for relaxation to promote stress management and mitigate against the risk of adverse cognitive and behavioral effects in spaceflight-like isolated, confined, and extreme (ICE) environments. Expanding upon previous work which investigated the feasibility of nature-based sensory stimulation using VR to promote stress management and relaxation ([4]), this project will (1) optimize the VR-based sensory stimulation experience through the integration of additional immersive components (e.g. haptic cues, enhanced audio), to promote engagement and thereby facilitate therapeutic release; (2) incorporate non-intrusive physiological monitoring for the objective assessment of relaxation; (3) promote relaxation through the introduction of biofeedback (i.e. VR presentation altered based on physiological cues); and (4) compare the effectiveness of various aspects of the VR experience for producing relaxation (via monitoring of physiological stress reduction) and restoring attention (through the measurement of performance on a cognition task). We will also examine individual preferences for sensory stimulation scenario characteristics (e.g. scene content, duration). References: [1] Kanas N, Sandal G, Boyd JE, Gushin VI, Manzey D, North R, (...), Inoue N. (2009). Psychology and culture during long-duration space missions. Acta Astronautica, 64(7-8), 659-77. [2] Stuster J. (2011). Bold endeavors: Lessons from polar and space exploration. Naval Institute Press. [3] Holland AW. (2000). Psychology of spaceflight. Journal of Human Performance in Extreme Environments, 5(1), 4-20. [4] Anderson, A. P., Mayer, M. D., Fellows, A. M., Cowan, D. R., Hegel, M. T., & Buckey, J. C. (2017). Relaxation with immersive natural scenes presented using virtual reality. Aerospace medicine and human performance, 88(6), 520-526.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	Virtual Reality platforms offer tremendous promise as psychological support tools in conditions of prolonged isolation and confinement.
Task Progress:	This project aims to optimize and test Virtual Reality (VR) sensory presentation for behavioral health support in isolated, confined, and extreme (ICE) environments. The work will include psychophysiological monitoring and feedback, and multisensory display presentations (e.g., haptic/tactile stimulation, enhanced audio), and will be tested in laboratory and ICE analog environments. The first phase of this investigation involved the analysis of subjective feedback questionnaires and post-mission interviews collected from participating operational ICE environment volunteers who interacted with a standard VR platform on an informal basis. The purpose of this exploratory, opportunistic research was to assess preference for VR scenarios and to gather contextually-specific experiential data, with the goal of optimizing future VR presentation for maximum restorative impact. This work has shown immersive VR to be highly rated, with natural scene content and dynamic scenes involving people and animals perceived as restorative following long periods of isolation and confinement. Findings suggest that options for personalized customization of the VR experience are also highly desirable. We deployed a set of VR experiences for usage at the South Pole Station, and have collected data during the past two winter-over seasons. These experiences included several different VR scenarios which modulated one or more of four specific attributes of the VR experience: (1) scenario duration (short vs long); (2) sensory modality (visual only or visual haptic cues and enhanced audio); (3) scene context (city or nature scenes); and (4) scene dynamic presentation (fixed scenes or dynamically explorable scenes with motion). We also investigated differential effects of assigned vs. self-selected VR sessions. For both seasons, we collected subjective response data through self-reported mood and preference questionnaires, as well as objective physiological responses to VR experience interactions during the second half of winter-over (approximately July-October), to assess the emotional and psychological impacts of various platform configurations. In the next several months, we will be conducting laboratory testing which will expand upon our analog work and incorporate an investigation of the impact of various VR experiences (including the introduction of biofeedback) on cognitive performance. We look forward to sharing our results soon!
Bibliography Type:	Description: (Last Updated: 03/18/2026)
Abstracts for Journals and Proceedings	Stankovic A, Thoolen S, Watson JM, Olsen E, Rodenberg H, Buckey JC, Strangman G. "VR In ICE: Quantification of response to virtual reality-based sensory stimulation for relaxation and stress relief in ICE environments." 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. Abstracts. 2024 NASA Human Research Program Investigators' Workshop, Galveston, TX, February 13-16, 2024. , Feb-2024